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SCHOOL FUTURE TECH PROGRAMM

Faculty Name :

Course Duration	: 5 Months (150 Days)
Classes	: Offline & Online
Daily Teaching	: 3 Hours
Total Hours	: 300 Hours

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LEVEL 1 – JUNIOR INNOVATORS (5th–6th Standard)

MODULE 1 – COMPUTER BASICS (Classes 1–20)

Class 7 — Computer Software – System Software & Application Software (Complete Notes)

Complete Theory & Practical Notes

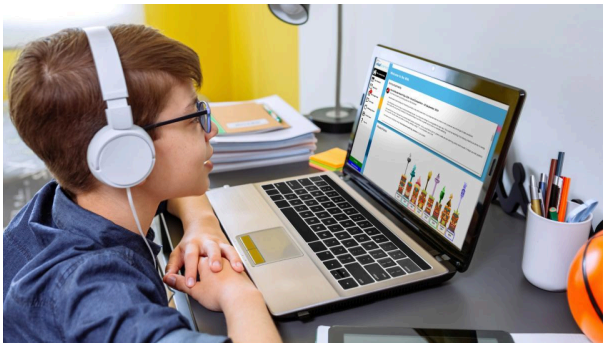
Learning Objectives

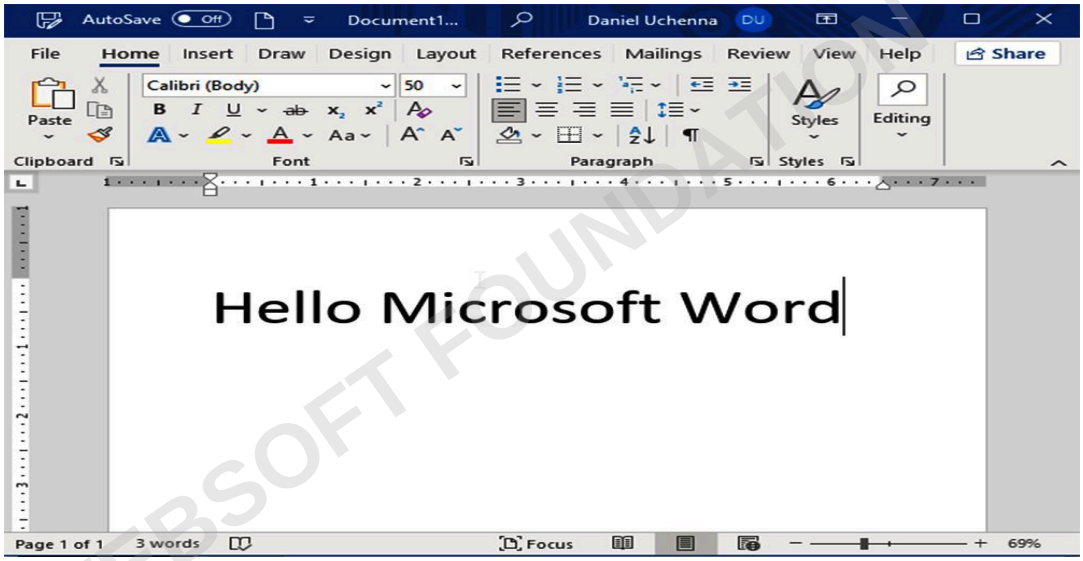
By the end of this class, students will be able to:-

- Understand what software is.
- Explain the difference between Hardware and Software.
- Identify System Software and Application Software.
- Name common examples of software.
- Understand why software is important.
- Create an interactive Scratch project to learn software types.

What is Computer Software?







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Definition:-

Software is a collection of programs and instructions that tells the computer ***what to do.***

- Without software, a computer cannot perform any task.

Simple Example:-

- You tell your friend to draw a house.
- Your instructions help your friend draw.

Similarly,

Software gives instructions to the computer.

Hardware vs Software

Hardware	Software
Physical parts of a computer	Programs and instructions
Can be touched	Cannot be touched
Examples: Monitor, Keyboard, Mouse	***Examples:*** Windows, Paint, MS Word

Easy Memory Trick

Hardware = What you can Touch

Software = What you can Use but cannot Touch

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Types of Software

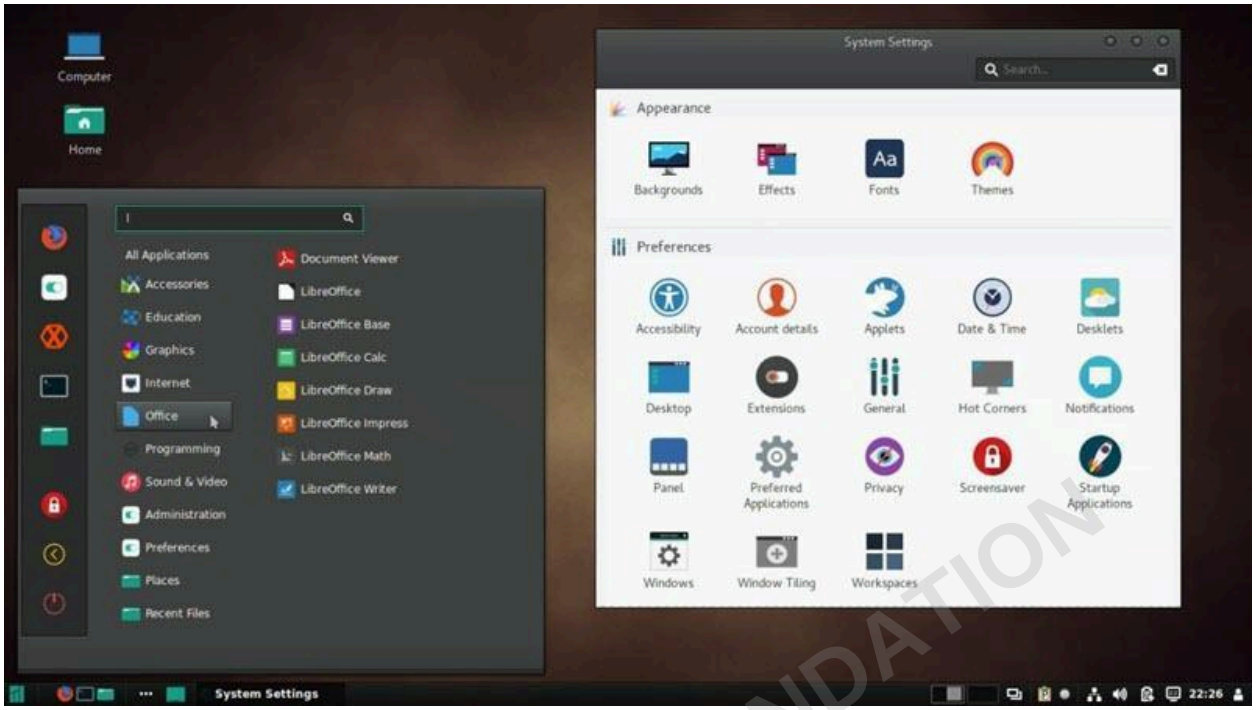
There are ***two main types*** of software:

1. System Software
2. Application Software

1. System Software







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What is System Software?

System Software controls the computer and helps all hardware and applications work properly.

It starts the computer and manages all its resources.

Functions

- Starts the computer
- Controls hardware
- Runs other software
- Manages files
- Controls memory

Examples

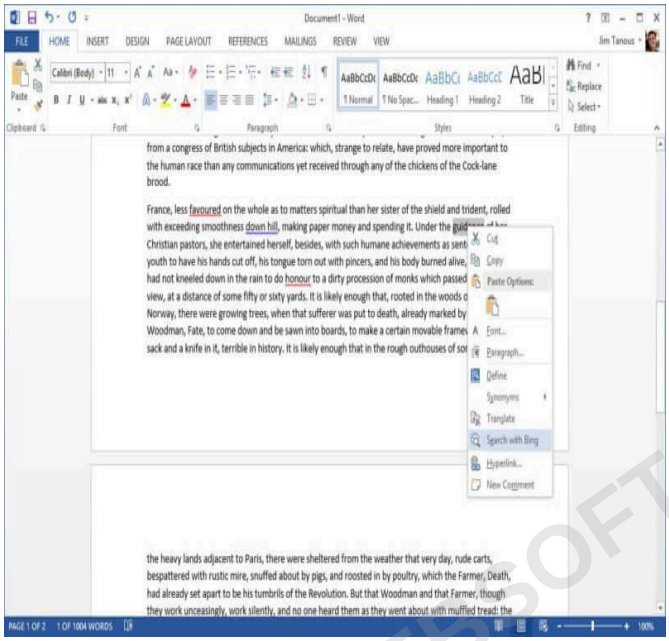
- Microsoft Windows
- Linux
- macOS
- Android

- iOS

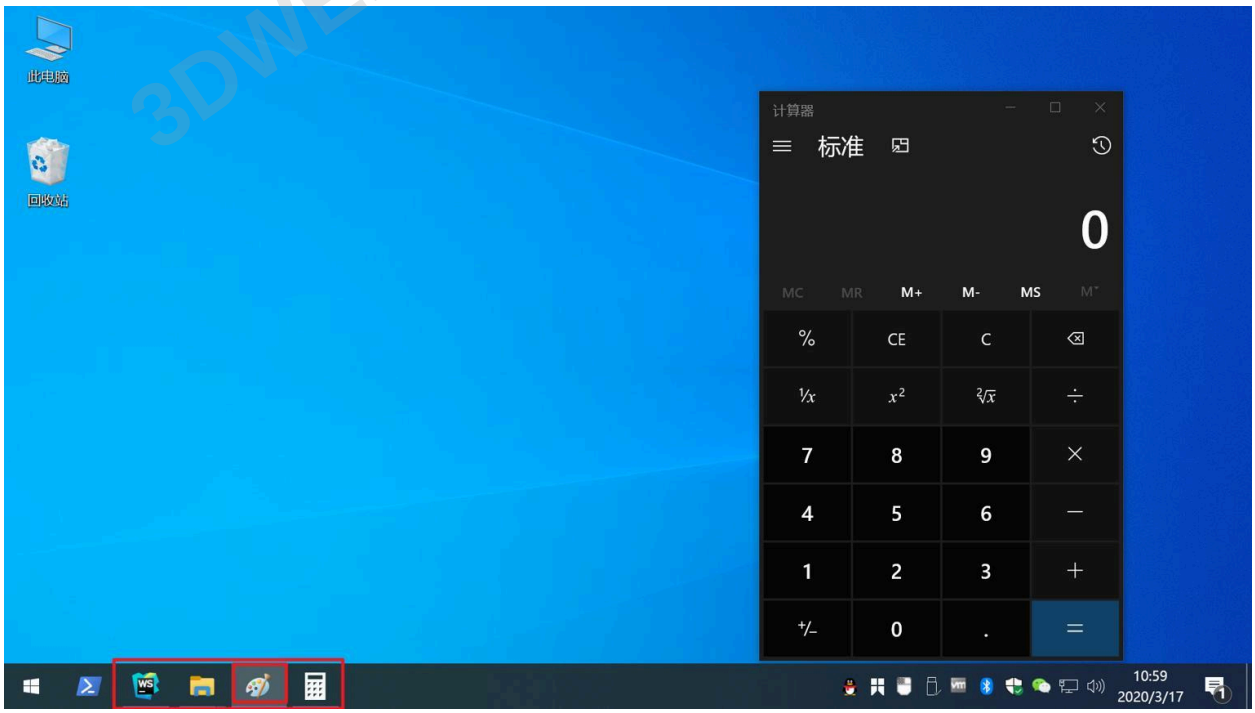
Fun Fact

Without an operating system, the computer cannot be used normally.

2. Application Software







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What is Application Software?

Application Software helps users perform specific tasks.

Uses

- Writing documents
- Drawing pictures
- Browsing the Internet
- Playing games
- Watching videos
- Learning online

Examples

- Microsoft Word
- Paint
- Calculator
- Google Chrome
- Scratch
- VLC Media Player

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Comparison: System Software vs Application Software

Feature	System Software	Application Software
Purpose	Runs the computer	Helps users perform tasks
Starts with Computer	Yes	No
Required	Yes	Optional
Example	Windows	Paint

Types of Software

Main Types of Computer Software

Software is broadly divided into two categories.

- **Application Software**
- **System Software**

Examples Around Us

Activity	Software Used
Typing homework	Microsoft Word
Drawing pictures	Paint
Browsing websites	Google Chrome
Learning coding	Scratch
Listening to music	VLC Media Player

Classroom Practical Activity

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Activity 1 – Identify the Software

The teacher opens different programs on the computer.

Students identify whether each is ***System Software or Application Software.***

Software	Type
Windows	System Software
Paint	Application Software
Scratch	Application Software
Calculator	Application Software
Android	System Software

Activity 2 – Match the Software

Draw two columns.

Software	Category
Windows	System Software
Linux	System Software
Paint	Application Software
Scratch	Application Software
Chrome	Application Software

Activity 3 – Open an Application

Students:-

1. Start the computer.
2. Open ***Paint.***
3. Draw a simple house or tree.
4. Save the drawing.
5. Close the application.

Learning Outcome: -Students understand that an operating system (system software) allows applications like Paint to run.

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Practical Coding Activity (Scratch)

Project Title

"Software Explorer"

Objective

Students create an interactive Scratch project where clicking software icons explains their type and use.

Step 1

Open **Scratch**.

Create a new project.

Delete the default Cat sprite.

Step 2

Add Sprites

- Windows Logo
- Paint Palette
- Scratch Logo
- Calculator
- Web Browser

Choose a computer desktop background.

Step 3 – Programming

Green Flag

when green flag clicked
say "Welcome! Click a software icon to learn about it." for 3 seconds

Windows Sprite

when this sprite clicked
say "I am Windows. I am System Software." for 4 seconds

Paint Sprite

when this sprite clicked
say "I am Paint. I will help you draw pictures." for 4 seconds

Scratch Sprite

when this sprite clicked
say "I am Scratch. I will help you learn coding." for 4 seconds

Calculator Sprite

when this sprite clicked
say "I help you solve math problems." for 4 seconds

Browser Sprite

when this sprite clicked

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say "I help you browse websites on the Internet." for 4 seconds

Advanced Scratch Challenge

Create a quiz using a ***Score*** variable.

Example:-

ask "Which software starts the computer?" and wait

If answer =

Windows

Then

change Score by 1
say "Excellent!" for 2 seconds

Else

say "Try Again!" for 2 seconds

Add More Questions

- Which software is used for drawing?
- Which software helps you type documents?
- Which software is used for web browsing?
- Which software is used for learning coding?
- Which software controls the computer?

Display the final score after all questions.

AI Connection

Discuss with students:-

- Which software lets you chat with an AI assistant?
- Which software is needed before AI applications can run?
- Can application software work without system software?
- Why is an operating system important for AI and games?

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Fun Quiz:-

1. What is software?

- A) A computer table
- ***B) A set of programs that tells the computer what to do***
- C) A monitor

2. Which is System Software?

- A) Paint
- ***B) Windows***
- C) Scratch

3. Which is Application Software?

- A) Linux
- B) Android
- ***C) Microsoft Word***

4. Which software is used for drawing?

- A) Chrome
- ***B) Paint***
- C) Windows

5. Which software is used to browse the Internet?

- A) Scratch
- B) Calculator
- ***C) Google Chrome***

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Homework:-

1. Draw a chart showing **Hardware** and **Software.**
2. List **5 examples of System Software** and **5 examples of Application Software**.
3. Open **Paint** or another drawing application at home and create a colorful picture.
4. Create a table comparing **System Software** and **Application Software.**
5. Improve your Scratch project by adding:
    - Sound effects

- Animated sprites
- A score counter
- A "Congratulations!" message after the quiz.

Learning Outcomes

After completing this lesson, students will be able to:-

- Define ***Computer Software.***
- Explain the difference between ***Hardware*** and ***Software.***
- Identify ***System Software*** and ***Application Software*** with examples.
- Understand the role of an operating system in running a computer.
- Use common application software such as Paint and Scratch.
- Build an interactive Scratch project that teaches software concepts in a fun and engaging way.

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